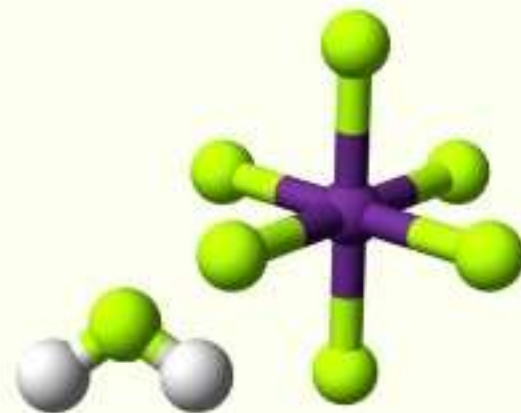
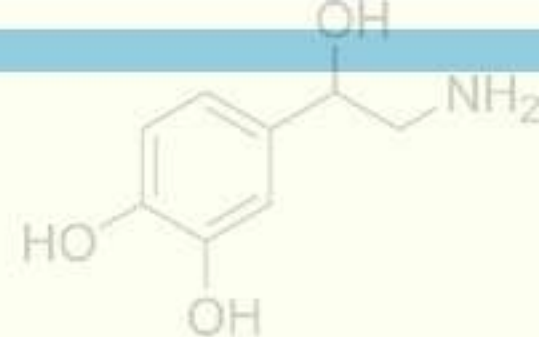


CHEMISTRY

METALS AND NON-METALS

MODULE - 7





MATTER

PURE MATTER

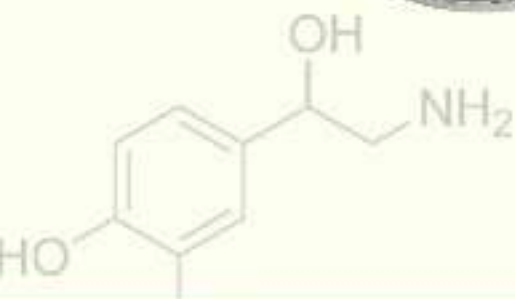
IMPURE MATTER

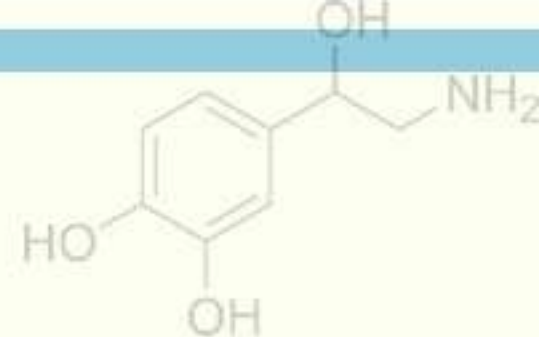
ELEMENT

COMPOUND

**HOMOGENOUS
MATTER**

**HETEROGENOUS
MATTER**





ELEMENT

METALS



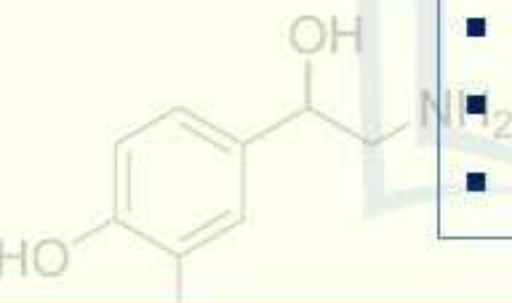
NON METALS



METALLOIDS



- Metals are usually solids
- Metalloids are solid in structure too at room temperature.
- But, non metals can exist in solid, liquid and gaseous states



DISTRIBUTION OF ELEMENTS IN THE PERIODIC TABLE

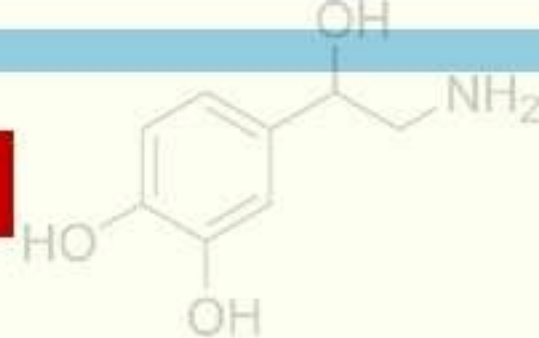
HO																		18 8A																	
1 1A												13 3A		14 4A		15 5A		16 6A		17 7A		18 8A													
H 1.00794 HYDROGEN		2 2A												B 10.811 BORON		C 12.011 CARBON		N 14.007 NITROGEN		O 15.999 OXYGEN		F 18.998 FLUORINE		He 4.0026 HELIUM											
		METALS										METALLOIDS				NONMETALS																			
3 Li 6.941 LITHIUM		4 Be 9.012 BERYLLIUM												13 Al 26.982 ALUMINUM		14 Si 28.086 SILICON		15 P 30.974 PHOSPHORUS		16 S 32.06 SULFUR		17 Cl 35.45 CHLORINE		18 Ar 39.948 ARGON											
11 Na 22.990 SODIUM		12 Mg 24.305 MAGNESIUM		3B		4B		5B		6B		7B		8B		11B		12B																	
19 K 39.098 POTASSIUM		20 Ca 40.078 CALCIUM		21 Sc 44.956 SCANDIUM		22 Ti 47.88 TITANIUM		23 V 50.942 VANADIUM		24 Cr 52.00 CHROMIUM		25 Mn 54.938 MANGANESE		26 Fe 55.845 IRON		27 Co 58.933 COBALT		28 Ni 58.69 NICKEL		29 Cu 63.546 COPPER		30 Zn 65.38 ZINC		31 Ga 69.723 GALLIUM		32 Ge 72.64 GERMANIUM		33 As 74.922 ARSENIC		34 Se 78.96 SELENIUM		35 Br 79.904 BROMINE		36 Kr 83.80 KRYPTON	
37 Rb 85.468 RUBIDIUM		38 Sr 87.62 STRONTIUM		39 Y 88.906 YTTRIUM		40 Zr 91.224 ZIRCONIUM		41 Nb 92.906 NIOBIUM		42 Mo 95.94 MOLYBDENUM		43 Tc 98.906 TECHNETIUM		44 Ru 101.07 RUTHENIUM		45 Rh 102.91 RHODIUM		46 Pd 106.42 PALLADIUM		47 Ag 107.87 SILVER		48 Cd 112.41 CADMIUM		49 In 114.82 INDIUM		50 Sn 118.71 TIN		51 Sb 121.76 ANTIMONY		52 Te 127.6 TELLURIUM		53 I 126.91 IODINE		54 Xe 131.29 XENON	
55 Cs 132.91 CESIUM		56 Ba 137.33 BARIUM		57-71 La-Lu LANTHANIDES		72 Hf 178.49 HAFNIUM		73 Ta 180.95 TANTALUM		74 W 183.84 WOLYBDENUM		75 Re 186.21 RHENIUM		76 Os 190.23 OSMIUM		77 Ir 192.22 IRIDIUM		78 Pt 195.08 PLATINUM		79 Au 196.97 GOLD		80 Hg 200.59 MERCURY		81 Tl 204.38 THALLIUM		82 Pb 207.2 LEAD		83 Bi 208.98 BISMUTH		84 Po 209 POLONIUM		85 At 210 ASTATINE		86 Rn 222 RADON	
87 Fr 223 FRANCIUM		88 Ra 226 RADIUM		89-103 Ac-Lr ACTINIDES		104 Rf 261 RUFORMIUM		105 Db 262 DUBNIUM		106 Sg 266 SEABORGIUM		107 Bh 264 BOHRIUM		108 Hs 277 HASSIUM		109 Mt 268 MEITNERIUM		110 Ds 271 DARMSTADTIUM		111 Rg 272 ROSGOLDIUM		112 Cn 285 COCHIN		113 Uut 288 UNUNTRIUM		114 Uuq 289 UNUNQUADIUM		115 Uup 291 UNUNPENTIUM		116 Uuh 293 UNUNHEXADIUM		117 Uus 294 UNUNSEPTIUM		118 Uuo 296 UNUNOCTADIUM	

57 La 138.905 LANTHANUM	58 Ce 140.12 CELESIUM	59 Pr 140.908 PRASEODYMIUM	60 Nd 144.24 NEODYMIUM	61 Pm 144.913 PROMETHIUM	62 Sm 150.36 SAMARIUM	63 Eu 151.964 EUROPEIUM	64 Gd 157.25 GADOLINIUM	65 Tb 158.925 TERBIUM	66 Dy 162.50 DYSPROSIUM	67 Ho 164.930 HOLMIUM	68 Er 167.259 ERBIUM	69 Tm 168.930 THULIUM	70 Yb 173.054 YTERBIUM	71 Lu 174.967 LUTETIUM
89 Ac 227.033 ACTINIUM	90 Th 232.038 THORIUM	91 Pa 231.036 PROTACTINIUM	92 U 238.029 URANIUM	93 Np 237.048 NEPTUNIUM	94 Pu 244.064 PLUTONIUM	95 Am 243.061 AMERICIUM	96 Cm 247.070 CURIUM	97 Bk 247.070 BERKELIUM	98 Cf 251.083 CALIFORNIUM	99 Es 252.083 EINSTEINIUM	100 Fm 257.103 FERMIUM	101 Md 258.103 MEITNERIUM	102 No 259.108 NORBOLTIUM	103 Lr 260.105 LAWRENCIUM





COMPARISON OF PHYSICAL CHARACTERISTICS



1

LUSTRE

METALS



NONMETALS



- Exceptional non metals with lustre – iodine and diamond.
- Diamond is an allotrope of carbon.





COMPARISON OF PHYSICAL CHARACTERISTICS

2

HARDNESS

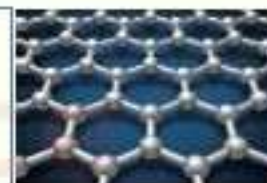
METALS



NONMETALS

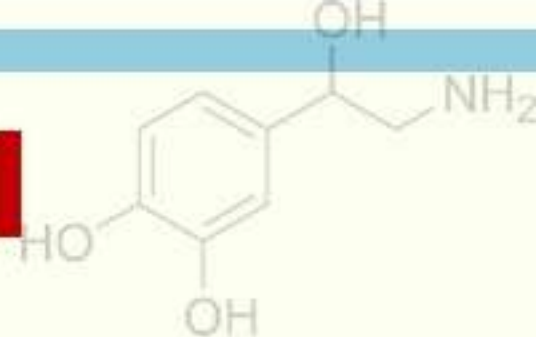


Diamond and Graphene are allotropes of carbon which are extremely hard.





COMPARISON OF PHYSICAL CHARACTERISTICS



3

MALLEABILITY

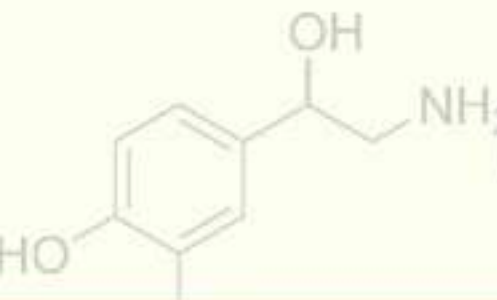
METALS



NONMETALS



Potassium and sodium are one of the least malleable metals because of their softness





COMPARISON OF PHYSICAL CHARACTERISTICS

4

DUCTILITY

METALS



NONMETALS



Potassium and sodium are one of the least ductile metals because of their softness.





COMPARISON OF PHYSICAL CHARACTERISTICS

5

THERMAL CONDUCTIVITY

METALS



NONMETALS



- Except graphite no non metals are good conductors of heat.
- Lead and mercury are poor conductors of heat.





COMPARISON OF PHYSICAL CHARACTERISTICS

6

ELECTRICAL CONDUCTIVITY

METALS



NONMETALS



- Silver is the best conductor of electricity.
- Graphite has good thermal and electrical conductivity.





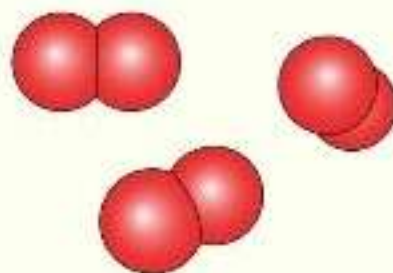
CHEMICAL PROPERTIES OF METALS

WHAT HAPPENS WHEN A METAL BURNS ?



Mg

+



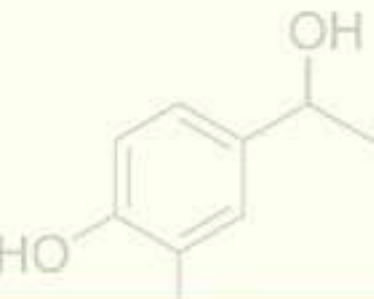
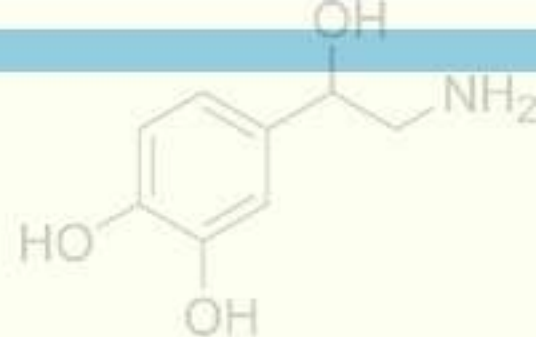
O₂

=



MgO

Metals react with oxygen(present in the flame) to form metal oxides





METALS GIVE OUT COLOURFUL FLAMES – EASY IDENTIFICATION



- Many metals burn with a characteristic flame.
- This nature is also exploited in identifying a particular metal in a compound.





DO ALL METALS REACT WITH OXYGEN THE SAME WAY?



Sodium

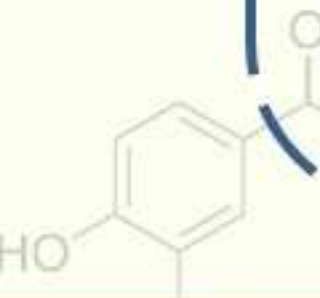


Copper

- All metals do not react with oxygen the same way.
- Some metals are very reactive and catch fire in the air;
- While others do not catch fire even if they are strongly heated.



THE METAL REACTIVITY SERIES





THE CHEMICAL NATURE OF THE METAL OXIDES

BASIC OXIDES

The oxides of metals are basic, they can react with acid and produce salt

AMPHOTERIC OXIDES

Aluminium and zinc oxides are amphoteric, they can react with both acids and base.

Potassium and sodium produces basic oxides which are soluble in water, these are called alkalis





REACTION OF METALS IN WATER

HIGHLY REACTIVE

- Potassium (cold water)
- Sodium (cold water)
- Lithium (cold water)
- Calcium (cold water)



MODERATELY REACTIVE

- Magnesium (hot water)
- Aluminum (steam)
- Zinc (steam)
- Iron (steam)



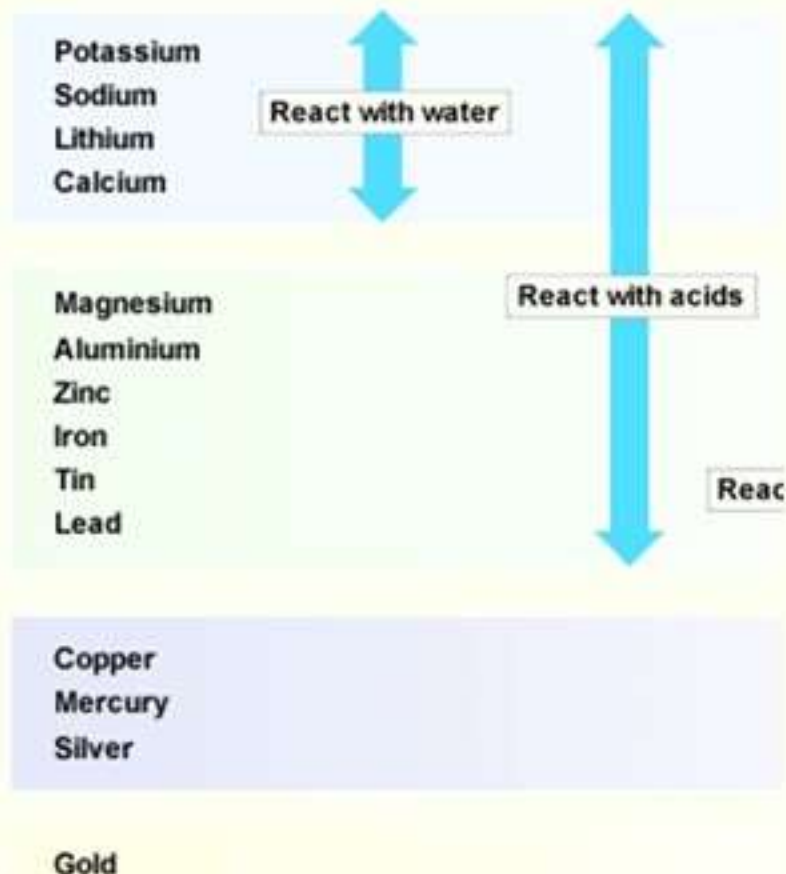
NONREACTIVE

- Lead
- Copper
- Silver
- gold





REACTION OF METALS WITH ACIDS



- Metals react with acids to give salt and hydrogen gas.
- When metals react to nitric acid, it forms salt and water, because the hydrogen is oxidised by nitrogen.
- Copper, mercury, silver and gold do not react with metals.





NOBLE METALS – NONREACTIVE TO ACIDS



Copper



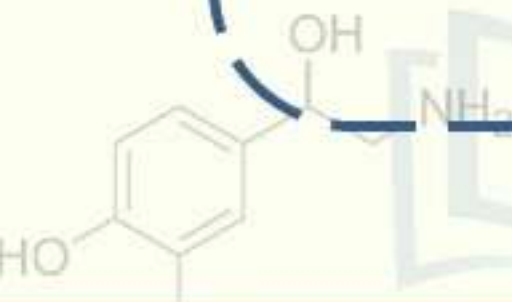
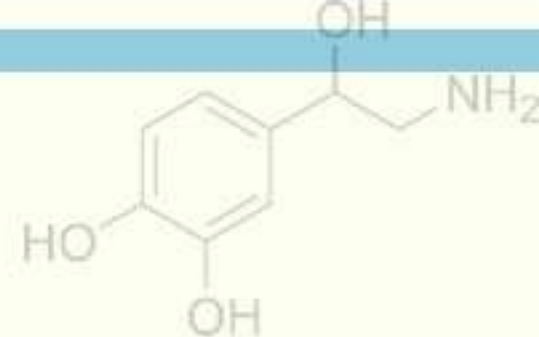
Silver



Gold



Platinum

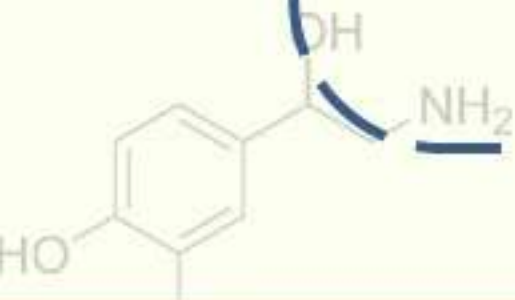
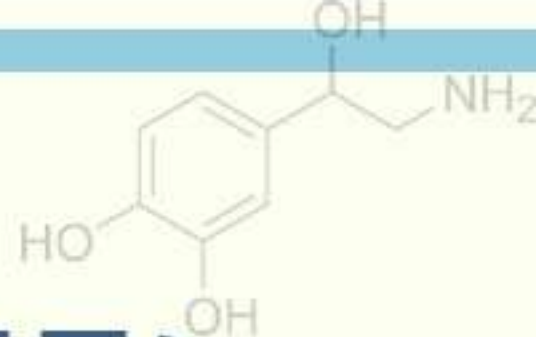




AQUA REGIA – CAN DISSOLVE GOLD !

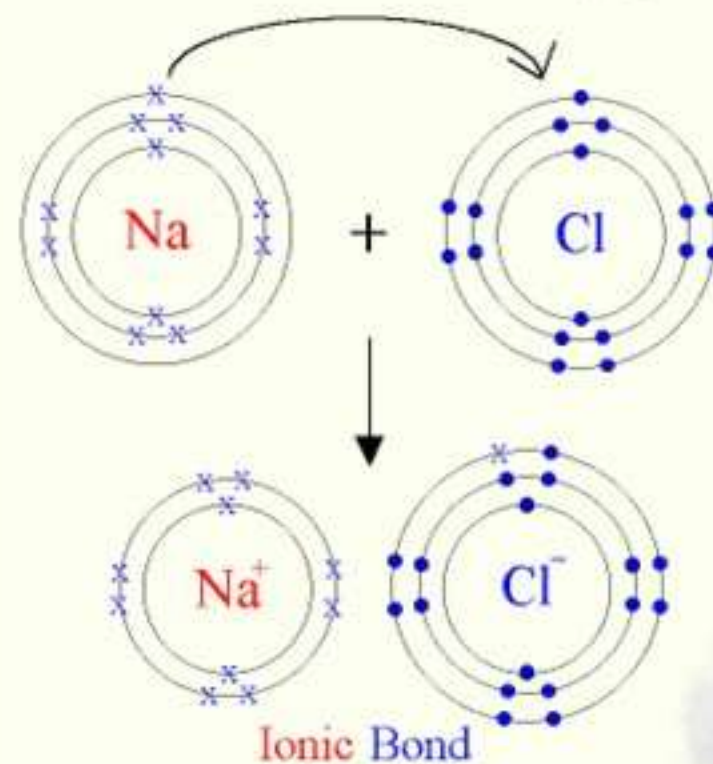
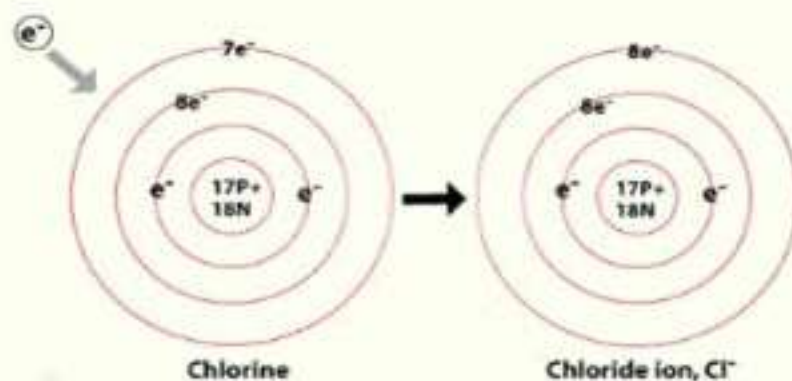
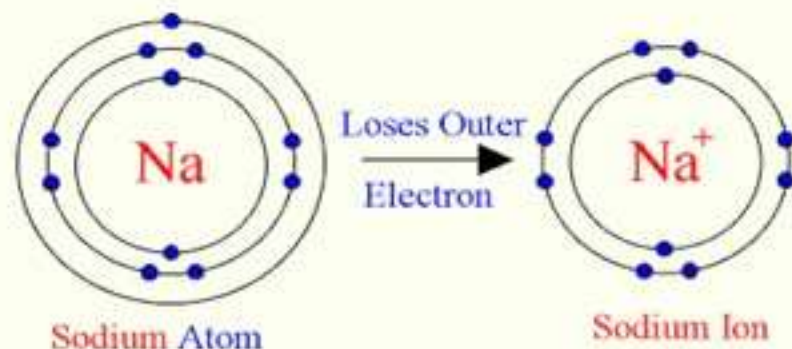


- **Aqua regia** or the royal water can dissolve noble metals like gold and platinum.
- It is a mixture of 1 part nitric acid and 3 parts hydrochloric acid.
- Aqua regia can't attack silver though!





HOW DO METALS REACT WITH NONMETALS ?



- Metals are found in a combined state.
- They can be extracted into pure metals



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